MIL-B-3137
16 FEBRUARY 1950
SUPERSEDING
Army 4-1016B
July 1, 1936

MILITARY SPECIFICATION

BENZENE, NITRATION GRADE

This specification was approved by the Departments of the Army, the Navy, and the Air Force for use of procurement services of the respective Departments.

1. SCOPE AND CLASSIFICATION

1.1 **Scope.**—This specification covers one grade of nitrobenzene suitable for use in the manufacture of nitrobenzene and as a solvent for purification of tetryl.

2. APPLICABLE SPECIFICATIONS, AND OTHER PUBLICATIONS

2.1 **Specifications.**—The following specifications, of the issue in effect on date of invitation for bids, form a part of this specification:

FEDERAL SPECIFICATION

TT-P-141—Paint, Varnish, Lacquer and Related Materials; Methods of Inspection, Sampling, and Testing.

U. S. ARMY SPECIFICATIONS

50-0-1—General Specification for Ammunition except Small Arms Ammunition.[1]

100-2—Standard Specification for Marking Shipments by Contractors.[1]

(*Army.*—Copies of specifications should be obtained from the procuring agency or as directed by that agency. Both the title and identifying number or symbol should be stipulated when requesting copies.)

NAVY DEPARTMENT SPECIFICATION

General Specification for Inspection of Material.[2]

(*Navy.*—Copies of Federal, Military, and Navy Department specifications may be obtained upon application to the Bureau of Supplies and Accounts, Navy Department, Washington 25, D. C., except that activities of the Armed Forces should make application to the Supply Officer in Command, Naval Supply Center, Norfolk 11, Va. Both the title and identifying number or symbol should be stipulated when requesting copies.)

(*Air Force.*—Copies of Federal, Military, and U. S. Air Force specifications may be obtained upon application to the Commanding General, Air Matériel Command, Wright-Patterson Air Force Base, Dayton, Ohio. Both the title and identifying number or symbol should be stipulated when requesting copies.)

2.2 **Other publications.**—The following publications, of the issue in effect on date of invitations for bids, form a part of this specification:

BUREAU OF SUPPLIES AND ACCOUNTS PUBLICATION

Navy Shipment Marking Handbook.[2]

(Copies of the Navy Shipment Marking Handbook may be obtained upon application to the Bureau of Supplies and Accounts, Department of the Navy, Washington 25, D. C., except that activities of the Armed Forces should make application to the Commanding Officer, Naval Supply Center, Norfolk 11, Va.)

INTERSTATE COMMERCE COMMISSION REGULATIONS

Regulations for Transportation of Explosives and Other Dangerous Articles, Etc.

(Information as to the availability of Interstate Commerce Commission Regulations for Transportation of Explosives and Other Dangerous Articles, etc., may be obtained from the Interstate Commerce Commission, Washington 25, D.C.)

3. REQUIREMENTS

3.1 **Specific gravity.**—Specific gravity shall be 0.8820 to 0.8860 at 15.56° C./15.56° C.

3.2 **Color.**—The color shall be not darker than a solution of 0.0030 gm. of $K_2Cr_2O_7$ in 1 liter of water. (See 4.4.)

3.3 **Distillation range.**—At 760 mm. pressure, the total distillation range for any sample shall be not more than 1° C., including the temperature of 80.1° C.

[1] Applicable only to Army purchases.
[2] Applicable only to Navy purchases.

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3.4 Solidifying point.—The solidifying point on anhydrous basis shall not be lower than 4.85° C.

3.5 Acidity.—There shall be no evidence of acidity in the benzene when tested as specified in 4.3.

3.6 Sulfur content.—There shall be no evidence of H_2S (hydrogen sulfide) or SO_2 (sulfur dioxide) in the benzene when tested as specified in 4.3.

3.7 Copper corrosion.—The copper test strip shall show no iridescence, gray or black deposit, or discoloration when in contact with the benzene as specified in 4.3.

3.8 Color of acid wash.—The color of the acid wash shall be not darker than No. 2 color standard. (See 4.3.)

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Sampling.—Samples shall be taken in accordance with the method specified in Specification TT-P-141.

4.2 Inspection.—Unless otherwise specified, benzene shall be inspected at the point of delivery. Unless otherwise specified, inspection shall be made in accordance with Specification 50-C-1.

4.3 Tests.—Unless otherwise specified, all tests shall be made in accordance with Specification TT-P-141.

Test	Method No.	ASTM equivalent
Acidity	525.5	D847
Specific gravity	418.3	D891
Distillation	430.3	D850
Solidifying point	542.1	D852
Color of acid wash	532.2	D848
Sulfur content	531.1	D853
Copper corrosion	510.1	D849

4.4 Color.—The color shall be determined by means of 50-ml. matched Nessler tubes making a visual comparison of the sample with a solution of 0.0030 gm. of reagent grade potassium dichromate in 1 liter of distilled water.

5. PREPARATION FOR DELIVERY

5.1 Packing.—Unless otherwise specified, nitration grade benzene shall be packed to insure acceptance by common or other carrier for safe transportation, at the lowest rate, to the point of delivery. All containers shall conform to the requirements of Interstate Commerce Commission Regulations for Transportation of Explosives and Other Dangerous Articles, etc.

5.2 Marking.—Each individual container shall be conspicuously marked as follows: "*Caution.* Volatile solvent—highly flammable. Avoid inhalation of fumes. Provide adequate ventilation in work areas." The applicable requirements of Interstate Commerce Commission Regulations for Transportation of Explosives and Other Dangerous Articles, etc., shall apply. In addition, shipments for the Army shall be marked in accordance with Specification 100-2; for the Navy, in accordance with the Navy Shipment Marking Handbook.

6. NOTES

6.1 Ordering data.—Invitations for bids and contracts or orders should specify the title, number, and date of this specification.

6.2 Benzene vapor has toxic properties and is also highly flammable. Precautions should be taken to avoid fire hazards and explosions. When used indoors, good ventilation is necessary.

6.3 Industrial pure and industrial 90 grades of benzene, formerly covered by Specification 4-1016B (listed as grades B and C), are now covered by Specification VV-B-231.

Notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Custodian:
Army—Ordnance Department.
Other interest:
Navy—OMS.

B-41650

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

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